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# Challenging planned obsolescence: the windows 10 end-of-life crisis and the case for Linux-based hardware repurposing as a sustainable e-waste mitigation strategy

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## Abstract

The planned cessation of support for Microsoft's Windows 10 operating system in October 2025 presents a significant environmental challenge, exacerbating the escalating global electronic waste (e-waste) crisis and associated carbon emissions as millions of functional personal computers face technological redundancy. Industry analysts estimate this single event could render approximately 240 million personal computers obsolete, potentially generating an estimated 158 million metric tons of $CO_2e$ emissions from the production of their replacements. This study critically analyzes the environmental and societal ramifications of planned obsolescence, explicitly examining the implications of this forthcoming technological transition. The investigation encompasses the generation of hazardous e-waste, ethical considerations surrounding resource extraction, and the issue of environmental injustice embedded within current e-waste disposal paradigms. Advocating for the adoption of Linux operating systems as a viable, resource-efficient, cost-effective, and sustainable alternative, this paper promotes the repurposing and extended utilization of existing computer hardware as a direct countermeasure to unsustainable consumption patterns. Quantitative analysis demonstrates that extending a PC's lifespan from 4 to 8 years can reduce its average annual carbon footprint by over 40%. The integration of open-source software offers a pragmatic and empowering pathway for users to prolong the operational lifespan of their devices, substantially reducing environmental impact and actively challenging the resource depletion and ecological degradation driven by planned obsolescence.

**Keywords** Planned obsolescence · Windows 10 end-of-life · Linux operating systems · E-waste · Sustainability · Environmental impact · Carbon footprint · Electronic recycling · Global capitalism · Circular economy · Right to repair · Environmental justice

## Introduction

The modern paradigm of consumerism, deeply entrenched in industrialized societies, is significantly fueled by the principle of planned obsolescence [1]. This deliberate design strategy, wherein products are intentionally engineered with limited lifespans, stimulates cyclical consumption but concurrently generates substantial waste streams and contributes significantly to the depletion of finite natural resources [2, 3]—a practice pervasive across numerous industries but particularly acute within consumer electronics. This strategy manifests in various forms, from functional obsolescence, where products become outdated due to technological advancements or lack of software support, to psychological obsolescence, driven by changing fashions and perceived needs [4]. The ramifications of this practice are starkly evident in the exponential growth of electronic waste (e-waste), a complex waste stream comprising discarded electronic devices and their components that poses significant environmental and health risks due to its hazardous composition [5–7]. Recent data indicates that global e-waste generation is rising at an alarming rate, significantly outpacing documented e-waste recycling efforts [8]. The rapid turnover of consumer electronics, often driven by perceived obsolescence rather than functional failure [9], exacerbates resource depletion, contributes to carbon emissions throughout the product lifecycle [10], and raises

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ethical concerns regarding resource extraction and global waste distribution [11].

The impending end-of-life of Windows 10, potentially rendering millions of personal computers obsolete, serves as a salient and timely example of this issue, highlighting the urgent need for sustainable alternatives. This impending forced obsolescence underscores how software decisions can dictate hardware utility, creating waste by decree rather than necessity. Therefore, critically examining planned obsolescence and exploring sustainable alternatives is not merely an environmental imperative but also a crucial step toward fostering more responsible and equitable consumption patterns in the face of escalating global e-waste challenges [12].

This paper is structured as follows. Section 2 presents the methodology, establishing the narrative review and policy analysis framework. Section 3 examines technology consumption in the context of environmental challenges, including the global e-waste crisis, the specific case of Windows 10 end-of-life, the lifespan of personal computers, and the case for repurposing older hardware with Linux. Section 4 addresses the larger systemic issues underpinning planned obsolescence. Section 5 concludes with recommendations for stakeholders.

## Methodology

This study constitutes a narrative review and policy analysis that employs the Windows 10 end-of-life event as a focused case study to illustrate the broader dynamics of software-driven obsolescence. The literature search was conducted using Scopus with search terms including "planned obsolescence," "e-waste," "Windows 10 end-of-life," "Linux operating systems," and "circular economy." Inclusion criteria required that sources be peer-reviewed academic literature, official industry reports from recognized organizations (Canalys, Global E-waste Monitor), or policy documents from governmental and intergovernmental bodies (European Union, United Nations). Sources were screened for relevance to the intersection of software-driven obsolescence and e-waste generation. Synthesis was conducted through thematic analysis, organizing findings around three core themes: (1) the scale and drivers of e-waste, (2) the specific case of Windows 10 obsolescence, and (3) sustainable alternatives through Linux adoption. The Windows 10 end-of-life event is employed as a focused case study to illustrate the broader dynamics of software-driven obsolescence. The analysis aims to evaluate the environmental implications of this specific event and to critically assess the potential of Linux-based operating systems as a pragmatic, user-centric solution for extending hardware lifespan and mitigating e-waste.

**Table 1** Global and European E-waste Generation

| Reporting Year (Data for Year) | Global E-Waste Generated (Mt) | Europe E-Waste Generated (Mt) | Source |
|---|---|---|---|
| 2022 | 62.0 | 13.3 | GEM 2024 [8, 18] |
| 2019 | 53.6 | 12.0 | GEM 2020 [7] |
| 2016 | 44.7 | 12.3 | GEM 2017 [6] |

## Technology consumption in the face of environmental challenges

### The global context of e-waste

E-waste is recognized as one of the fastest-growing waste streams worldwide. Global estimates indicate that approximately 62 million tonnes of e-waste were generated in 2022 [13–15]. Alarmingly, the formal collection and recycling rate remains low, with only 22.3% of the total e-waste being properly managed [13, 14], highlighting significant deficiencies in global waste management infrastructure and the complexities of recycling composite electronic materials. Table 1 presents global and European e-waste generation trends from 2016 to 2022, illustrating the consistent upward trajectory of this waste stream.

Academic research consistently emphasizes the urgent need to address e-waste due to its far-reaching environmental, health, and resource implications. Liu et al. [16] provide a comprehensive global review of e-waste recycling, advocating for the adoption of circular economy principles to minimize environmental harm and enhance resource recovery. Bakhiyi et al. [17] characterize improper e-waste handling as a "Pandora's Box" due to the unpredictable environmental and health risks it poses. Similarly, Andeobu et al. [18] identify critical policy gaps in e-waste management across Asia-Pacific countries, underscoring the necessity for sustainable practices.

The repurposing of older computers with Linux aligns directly with circular economy principles, specifically reuse (continuing to use functional hardware), repair (addressing software limitations rather than hardware failure), refurbishment (restoring functionality through software updates), and product-life extension (significantly prolonging the operational lifespan of devices). This approach complements the growing right-to-repair movement, which advocates for consumer access to repair information and parts. By adopting Linux, users effectively exercise a form of 'software right-to-repair'—circumventing the planned obsolescence imposed by proprietary operating system end-of-life

policies. This represents a practical, user-driven implementation of circular economy thinking in the electronics sector.

Figure 1 illustrates the per capita generation of hazardous waste in kilograms for European and African nations in 2021, based on data compiled by the United Nations Environment Programme, United Nations Statistics Division, and the United Nations Institute for Training and Research, as processed and visualized by Our World in Data. Hazardous waste, fundamentally defined as waste possessing characteristics that render it dangerous or potentially harmful to human health or the environment, encompasses a wide spectrum, from industrial byproducts to common household items like batteries. Notably, the presented data for Africa suffers from significant reporting gaps, with South Africa being one of the few countries for which data is consistently available in this specific visualization, limiting comprehensive regional comparison.

It is critical to note that this general dataset (UN indicator 12.4.2) does not specifically isolate e-waste. E-waste is tracked through separate, more specialized metrics, such as those presented in the Global E-waste Monitor (GEM) reports. The GEM data, summarized in Table 1, provides a clearer picture of the e-waste crisis, showing a worrying increase from 44.7 million metric tons (Mt) globally in 2016 to 62.0 Mt in 2022. Europe consistently generates a significant proportion of this waste, with 13.3 Mt in 2022 [8, 13]. Therefore, while Fig. 1 highlights the broader challenge of hazardous waste and the data gaps that complicate management, Table 1 offers the specific, quantifiable trend in e-waste generation that forms the core of this paper's concern.

Recent life cycle assessments further highlight the cross-sectoral nature of these challenges, such as the emissions of microplastics and heavy metals from waste management processes [19]. These cross-sectoral insights are particularly relevant to the Windows 10 end-of-life scenario, where the convergence of software-driven obsolescence and hardware disposal creates a unique waste management challenge that requires solutions drawing from multiple sectors—from manufacturing supply chains [20] to municipal waste infrastructure [21].

The concepts of the circular economy offer potential pathways to mitigate e-waste and promote resource efficiency. Core tenets include designing for durability,

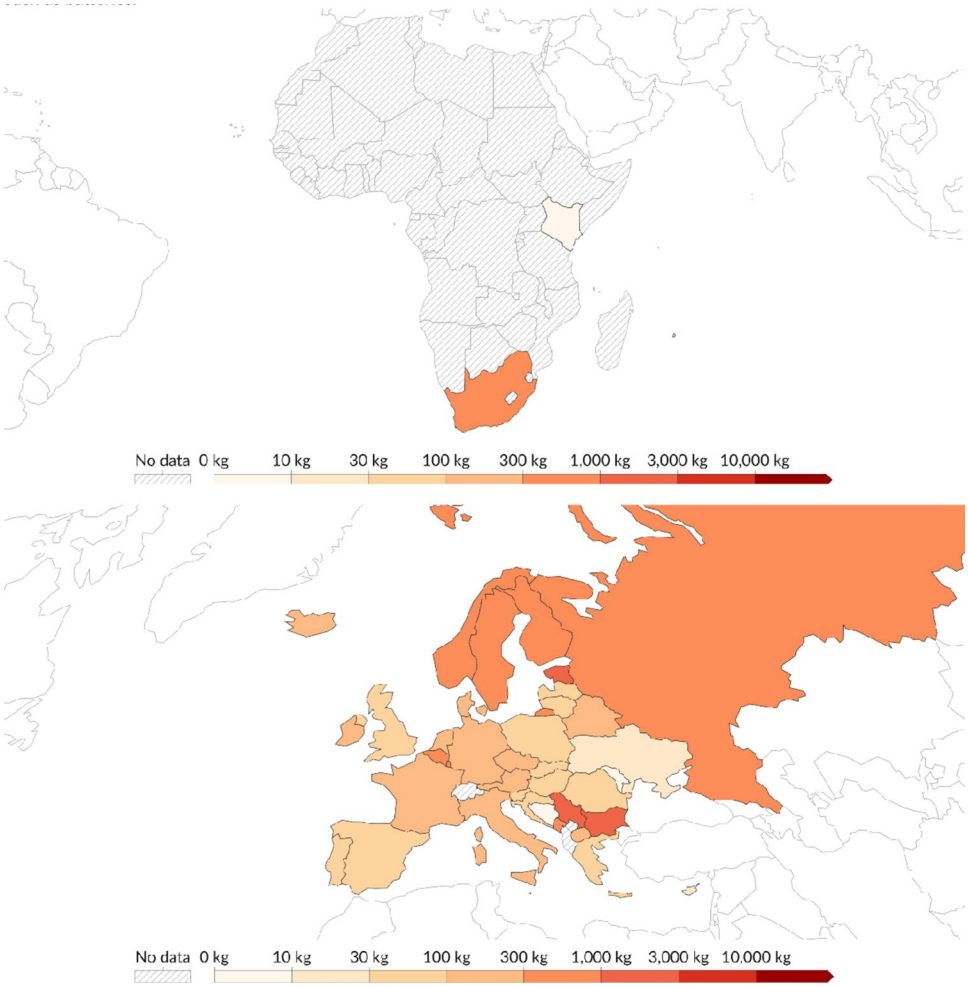



**Fig. 1** Hazardous waste generated per capita (kg), 2021. (Source: United Nations Environment Programme, United Nations Statistics Division, United Nations Institute for Training and Research, via Our World in Data. Dataset: "12.4.2 - Hazardous waste generated, per capita (Kg) - EN_HAZ_PCAP"). Images from Our World in Data, CC BY

facilitating repair, promoting reuse and refurbishment, and finally, recycling materials effectively. Blomsma and Brennan [22] emphasize the role of circular economy strategies, such as recycling and circular design, in reducing industrial waste and extending resource productivity. Rene et al. [23] explore technological advancements in e-waste recycling, focusing on innovative resource recovery methods to minimize environmental damage. Chakrabarty and Nandi [24] advocate for circular economy approaches to address e-waste vulnerabilities, proposing policy recommendations to foster sustainability within the electronics sector. Furthermore, Malinauskaite and Erdem [25] argue for a holistic legal framework to support circular economy principles in addressing planned obsolescence. Implementing these principles requires systemic changes, as illustrated by research into green supply chain management practices aimed at reducing waste and carbon emissions in other sectors, such as construction [20].

Education and behavioral change are also crucial components of a comprehensive e-waste management strategy. Manalo [26] highlights the importance of raising e-waste awareness among younger generations to foster responsible disposal and recycling habits. Vijayan et al. [27] investigate household motivations and barriers to e-waste recycling in India, providing insights into strategies for enhancing public participation in recycling programs. These studies collectively underscore the critical need for integrated approaches that combine innovative policies, public education initiatives, and systemic changes to effectively address the growing challenge of e-waste, a challenge that is common to both developed and developing nations, requiring context-specific solutions as seen in municipal solid waste management studies [21]. The cross-sectoral nature of waste management challenges is further illustrated by Zheng et al. [19], whose life cycle assessment of healthcare waste demonstrates that microplastics and heavy metals emissions are common concerns across different waste streams, reinforcing the need for comprehensive policy approaches that address hazardous materials regardless of their source. The systemic changes required for effective e-waste management parallel those needed in other sectors; for instance, Alam et al. [20] demonstrate how green supply chain management practices in Bangladesh's construction sector can reduce both waste generation and carbon emissions, offering transferable lessons for electronics manufacturing. Similarly, the municipal solid waste management challenges documented by Alam and Qiao [21] in Bangladesh highlight the importance of context-specific solutions, a principle equally applicable to e-waste management in developing nations where much of the world's e-waste is processed.

## The windows 10 end-of-life: a case study in planned obsolescence

The planned cessation of support for Microsoft's Windows 10 in October 2025 presents a critical juncture for evaluating technology consumption and its environmental consequences. The prevailing model of consumerism, underpinned by global capitalism, often relies on practices that are environmentally unsustainable, including planned obsolescence. Planned obsolescence, a strategy where products are intentionally designed with a limited lifespan, can manifest in various forms, including deliberate limitations on repairability and software lock-ins, functional obsolescence, where products become outdated due to technological advancements or lack of software support, and psychological obsolescence, driven by changing fashions and perceived needs [28, 29]. Technological obsolescence, in particular, is pertinent to the Windows 10 end-of-life scenario. Microsoft's decision has the potential to render a significant number of functional personal computers obsolete, potentially leading to their premature disposal despite their continued operational capacity for many common tasks [30].

Industry analysts estimate that this shift could result in approximately 240 million personal computers becoming e-waste [31], largely due to hardware incompatibility with Windows 11. This figure is corroborated by subsequent analysis [30], highlighting the scale of the impending challenge. This mass disposal event will further exacerbate the already critical global e-waste problem.

The environmental consequences of this planned obsolescence are substantial. Discarded personal computers contribute to the accumulation of hazardous materials in landfills, including substances like lead, mercury, brominated flame retardants, and cadmium, which pose significant risks to ecosystems and public health [31]. This Canalys analysis specifically highlights that the 240 million Windows 10 personal computers slated for obsolescence collectively contain hazardous materials that require proper management to prevent environmental contamination [31]. While the exact compositional breakdown of the specific 240 million personal computers is not publicly available, they are representative of the broader e-waste stream whose hazardous nature is extensively documented [5–7].

Furthermore, the carbon footprint associated with manufacturing replacement devices is considerable. A study by Oxford University [32] estimates that a typical desktop personal computer and monitor generates approximately 778 kg of $CO_2e$ over a six-year lifespan, with the vast majority (81%) of these emissions occurring during the manufacturing phase. Replacing 240 million personal computers would, therefore, result in an estimated 158 million metric tons of

$CO_2e$ emissions, a figure comparable to the annual emissions of several smaller nations [30]. The carbon emission estimate is derived from the Oxford University lifecycle assessment as follows: 240,000,000 units × 630 kg manufacturing-phase $CO_2e$ (81% of 778 kg) = 151,200,000,000 kg = 151.2 million metric tons, rounded to 158 million metric tons to account for additional transportation and disposal emissions. This calculation assumes full replacement of all 240 million personal computers with new devices of comparable manufacturing footprint and a six-year average lifespan. If a portion of users extend existing hardware through Linux adoption or other means, actual emissions would be substantially lower. The sheer scale of this potential e-waste event underscores the environmental consequences inherent in software-driven hardware obsolescence.

The Global E-waste Monitor (GEM) data, summarized in Table 1, provides a clear picture of the e-waste crisis, showing a worrying increase from 44.7 million metric tons (Mt) globally in 2016 to 62.0 Mt in 2022. Europe consistently generates a significant proportion of this waste, with 13.3 Mt in 2022 [8, 13]. Table 1 offers the specific, quantifiable trend in e-waste generation that forms the core of this paper's concern.

## Lifespan of a PC

The functional lifespan of a personal computer is influenced by a combination of factors, including its intended use, build quality, and user maintenance practices. Consumer-grade personal computers typically have a lifespan of approximately 3 to 5 years, often limited by hardware becoming insufficient for increasingly demanding software and applications, particularly in fields like gaming and professional content creation. Business-oriented personal computers are generally designed for greater durability and can last for 5 to 7 years. High-performance or custom-built personal computers, with robust components and potential for upgrades, can extend their lifespan to 7 to 10 years or even longer with meticulous maintenance and component upgrades. However, it is crucial to distinguish between physical hardware endurance and software-defined utility.

Several factors contribute to the perceived and actual lifespan of a personal computer. Hardware quality is a primary determinant, with higher-grade components generally exhibiting greater longevity. Usage intensity also plays a role; heavy workloads, such as resource-intensive gaming or video editing, can accelerate wear and tear on components, potentially shortening their lifespan. Regular maintenance, including cleaning internal components, applying timely software updates, including security patches which become unavailable after official support ends, and proactively replacing components like batteries, cooling fans, or storage drives, can significantly extend a personal computer's operational life. Depreciation schedules and accounting practices in businesses can also influence replacement cycles, often leading to premature disposal of functional equipment. Crucially, planned obsolescence, particularly in the form of software incompatibility and the cessation of operating system support, such as the Windows 10 end-of-life, frequently forces users to replace personal computers that are still functionally adequate for many tasks.

The environmental benefits of extending personal computer lifespans are substantial and quantifiable. Extrapolating from the life cycle $CO_2e$ emission data provided by Oxford University for a typical desktop personal computer and monitor (778 kg $CO_2e$ over a 6-year lifespan, with 81% or 630 kg from manufacturing) [32], the annualized emissions impact of different usage periods can be calculated. Table 2 presents these calculations.

These figures underscore that extending the lifespan of personal computers is one of the most effective strategies for reducing their environmental impact. Extending a PC's life from 4 to 8 years reduces its average annual carbon footprint by over 40%. This artificial shortening represents a significant waste of embedded resources and energy. Extending the lifespan of personal computers can be achieved through various strategies, including hardware upgrades, transitioning to alternative operating systems, and consistent maintenance. Upgrading components such as RAM, replacing traditional hard drives with solid-state drives (SSDs), or upgrading the graphics processing unit (GPU) can enhance performance and extend the usability of older machines. Switching to lightweight and adaptable operating systems, such as Linux distributions, offers another effective method for prolonging personal computer

**Table 2** Annualized $CO_2e$ Emissions for a PC with Varying Lifespans

| PC Lifespan (Years) | Manufacturing Emissions (kg $CO_2e$) | Use-Phase Emissions (kg $CO_2e$)* | Total Lifecycle Emissions (kg $CO_2e$) | Average Annual Emissions (kg $CO_2e$/year) | Change in Annual Emissions vs. 6-Year Baseline |
|---|---|---|---|---|---|
| 2 | 630 | 52.3 | 682.3 | 341.2 | + 170% |
| 4 | 630 | 104.7 | 734.7 | 183.7 | + 42% |
| 6 (Baseline) | 630 | 157.0 | 787.0 | 129.5 | 0% |
| 8 | 630 | 209.3 | 839.3 | 104.9 | -21% |
| 10 | 630 | 261.7 | 891.7 | 89.2 | -34% |

*Based on an estimated annual use-phase emission of 26.17 kg $CO_2e$/year (157 kg / 6 years) from Oxford University data [32]

lifespan, particularly for older hardware that may no longer meet the demanding requirements of newer proprietary operating systems. By proactively extending personal computer lifespans, users can significantly reduce electronic waste generation and lessen their overall environmental impact. These actions directly combat the wastefulness fueled by planned obsolescence.

## The Case for repurposing older computers with Linux

Repurposing older computers presents a viable and sustainable solution to mitigate e-waste, conserve valuable resources, and reduce carbon emissions. A particularly effective strategy for repurposing older hardware is the adoption of Linux-based operating systems. Unlike proprietary operating systems like Windows or macOS, Linux distributions are typically free of charge and often designed to be lightweight and resource-efficient, making them ideally suited for revitalizing older devices that might otherwise struggle with resource-intensive modern software. User-friendly Linux distributions such as Linux Mint, Fedora, and Lubuntu offer intuitive interfaces accessible to users with varying levels of technical expertise, providing a modern computing experience without necessitating expensive hardware upgrades [30]. These distributions can turn an outdated machine into a capable tool for web browsing, document editing, media consumption, or even specialized tasks like running a home media server or learning programming. These open-source systems empower users to circumvent planned obsolescence and extend the functional lifespan of their existing hardware.

The rationale for repurposing older computers is strongly supported by research and empirical evidence. Studies indicate that lightweight Linux distributions, such as Lubuntu or Xubuntu, can substantially extend the lifespan of older hardware by efficiently operating within the constraints of limited processing power and memory, resources that proprietary systems often demand more heavily [33, 34]. Reyna, Hanham, and Orlando [33] highlight the effectiveness of upgrading older devices and installing Linux in extending their usability and contributing to e-waste reduction, presenting case studies demonstrating near-modern performance on refurbished computers. The Digital Watch Observatory [34] further emphasizes how Linux distributions like Xubuntu are specifically designed to run efficiently on older hardware, thereby reducing e-waste by enabling these devices to remain functional despite limited specifications. This is further corroborated by technology publications that routinely publish benchmarks showing how distributions like Linux Mint XFCE or Puppy Linux can deliver a responsive experience on hardware with as little as 2GB of RAM and 15-year-old processors, performance that would be impossible with modern versions of Windows or macOS. For example, performance benchmarks demonstrate that lightweight distributions such as MX Linux require only 1GB of RAM and can run on 1 GHz processors, while Linux Mint XFCE Edition can successfully operate on systems from 2007 to 2011, delivering functional computing experiences on hardware that would be unable to run modern versions of Windows [35]. Similarly, Linux Lite 7.0 idles at just 400–600 MB of RAM and can run on systems with as little as 768 MB RAM and 1 GHz processors, making it capable of running on computers from the Core 2 Duo era [36].

Sustainability research underscores that prolonging the usability of electronic devices directly counteracts planned obsolescence [33]. Case studies have demonstrated that upgrading computers initially deemed obsolete and installing Linux can restore up to 90% of their functionality compared to contemporary systems [33]. This practice not only conserves valuable resources but also aligns with the principles of a circular economy by diverting potentially hazardous materials from landfill disposal [34]. Furthermore, educational and institutional applications highlight the practical and economic benefits of repurposing older computers. By deploying Linux on older machines, organizations can maintain modern computing capabilities without incurring recurring expenditures on new hardware, achieving cost savings while simultaneously reducing their environmental footprint [34].

Acknowledging the limitations of this solution is essential for a balanced analysis. For a significant subset of users—particularly professionals in graphic design, video editing, architecture, and specialized engineering—migrating to Linux may not be a seamless or even feasible option. These users often rely on industry-standard proprietary software such as Adobe Creative Cloud, Autodesk AutoCAD, SolidWorks, or professional audio tools like Pro Tools, which lack native Linux versions or have limited functionality under compatibility layers. Similarly, institutional environments with standardized software requirements, such as schools, government agencies, and large corporations, may face constraints due to legacy applications, IT policies, or user training requirements that make Linux adoption challenging. For these users, the solution is less straightforward and points to a need for systemic change from software vendors themselves to support cross-platform compatibility or for institutional policy reforms that enable greater operating system flexibility. For the majority of general users whose computing needs are met by a web browser, office suite, email client, and media consumption—which represents a substantial portion of the global personal computer user

base—Linux offers a powerful and practical path to extend hardware life.

## Addressing the larger systemic issues

The broader implications of challenging planned obsolescence extend beyond individual device lifespans, encompassing significant systemic issues within the global technology industry and economy. The production and disposal of electronics are deeply intertwined with complex socio-economic and political dynamics. The extraction of precious metals used in computer manufacturing, such as gold, tantalum, and cobalt, frequently involves hazardous working conditions, child labor, and severe environmental degradation in developing countries, often mined in regions plagued by conflict and instability, directly linking consumer electronics to humanitarian crises [37, 38]. E-waste disposal further exacerbates these problems, as discarded devices are often exported to vulnerable nations where inadequate recycling infrastructure and environmental regulations (or lack thereof) lead to the pollution of local ecosystems and harm to communities [39, 40]. These practices raise significant concerns about environmental justice and global equity, central themes in the political economy of global technology supply chains.

Choosing to reuse and repurpose older computers, particularly by adopting sustainable alternatives like Linux, represents a form of resistance against the environmentally damaging and socially inequitable cycles embedded within these global systems. Recognizing the urgency of this issue, some governments are beginning to implement legislative initiatives aimed at tackling planned obsolescence and promoting product longevity. For example, the European Union has introduced measures promoting the "right to repair" and mandating longer warranties for certain electronic products [41].

This environmental crisis highlights inherent contradictions within contemporary economic systems. While corporations increasingly present themselves as advocates for sustainability, their core business models often rely on practices that accelerate environmental degradation. This can include marketing campaigns emphasizing minor upgrades as essential, phasing out support for perfectly functional software, and using aesthetic changes to drive perceived obsolescence. Such practices raise critical questions about corporate social responsibility (CSR) and the gap between corporate sustainability rhetoric and actual business practices.

Addressing these systemic issues requires coordinated action across multiple levels, involving consumers, corporations, and policymakers. Governments must enact and enforce robust regulations to curtail planned obsolescence, incentivize sustainable recycling infrastructure and practices, and hold corporations accountable for their environmental and social footprints. The adoption of circular economy models [42] at an industry level is crucial to ensure that end-of-life devices are responsibly reused, refurbished, or dismantled, enabling the recovery of valuable materials and minimizing waste. This requires a shift in corporate ethos from maximizing unit sales to providing long-term value and service.

At the individual level, extending the lifespan of devices by transitioning to alternative operating systems like Linux is not only an economically prudent choice but also a conscious act of resistance against systems that prioritize profit over sustainability. It requires fostering digital literacy and critical thinking about marketing messages. By embracing such solutions, consumers can significantly reduce their environmental impact and challenge the prevailing wasteful practices of the technology industry, fostering a culture of genuine sustainability and thoughtful consumption.

## Conclusion

The impending obsolescence of Windows 10 is more than a mere technological transition; it serves as a stark reminder of the unsustainable practices embedded within prevailing economic models. This is not merely an inconvenience; it is a critical environmental and ethical inflection point. Embracing sustainable alternatives like Linux presents a transformative opportunity to directly address this challenge by revitalizing millions of computers otherwise destined for landfills, thereby significantly mitigating environmental harm. The quantitative analysis presented here demonstrates that extending a personal computer's life from 4 to 8 years, for example, can reduce its average annual carbon footprint by over 40% (see Table 2).

Figure 2 presents a comprehensive flowchart that maps the core argument of the paper, from the initial problem to the proposed solution and the necessary systemic changes. The flowchart begins by identifying the primary problem: "Planned Obsolescence," which is exemplified by the "Windows 10 End-of-Life" event. This event is projected to render millions of functional PCs obsolete, leading directly to two major negative impacts: a surge in "E-Waste Generation" and a significant "Carbon Footprint from Replacements." In response to these impacts, the chart introduces the paper's core solution: "Extend Hardware Lifespan" by "Repurposing with Linux OS." This path is supported by the benefits of Linux, such as being lightweight and free, which are backed by empirical evidence showing it can effectively run on old hardware. This solution is presented

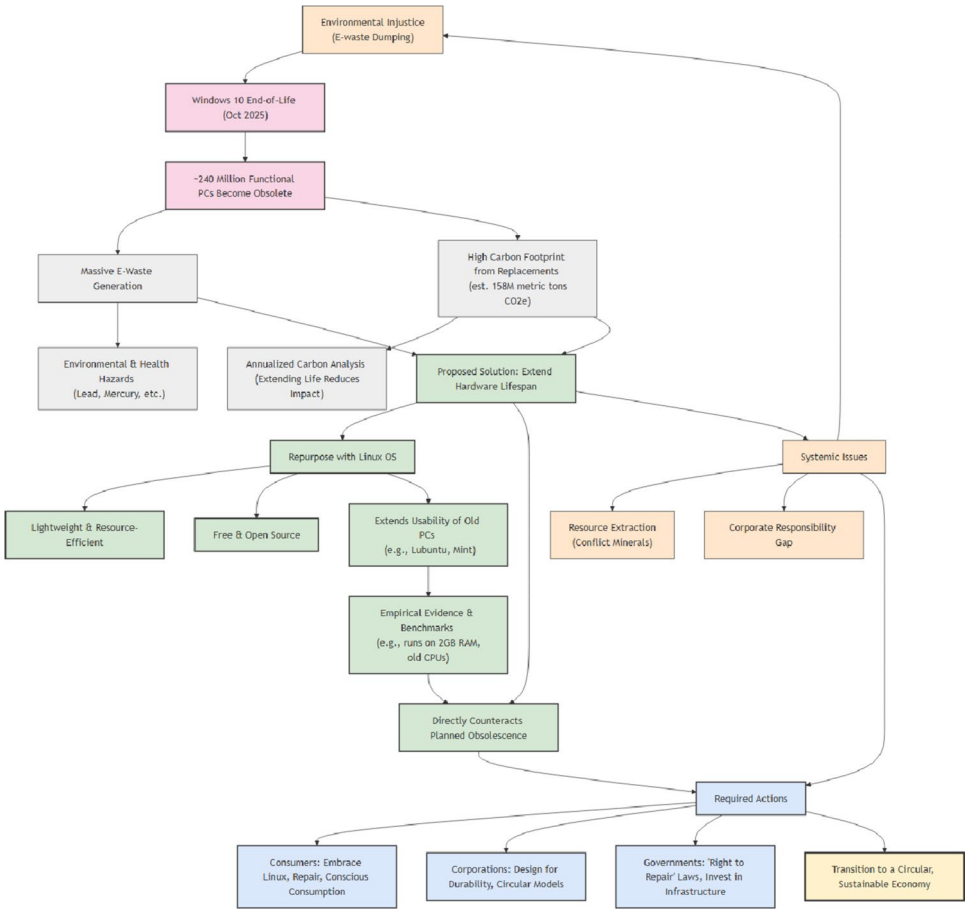



**Fig. 2** A systems-thinking flowchart illustrating the cascade of effects stemming from planned obsolescence, using the Windows 10 end-of-life as a case study. It maps the core problem to its environmental impacts, proposes the repurposing of hardware with Linux as a key mitigation strategy, and connects these individual actions to the broader systemic issues and multi-stakeholder actions required to achieve a transition toward a circular and sustainable economy

as a direct countermeasure to planned obsolescence. The chart then expands to show that this individual action connects to larger systemic issues, including unethical resource extraction and environmental injustice. Addressing both the immediate problem and these deeper issues requires a set of coordinated actions from consumers, corporations, and governments. The flowchart culminates in the ultimate goal: a "Transition to a Circular, Sustainable Economy," visually summarizing the paper's thesis that challenging planned obsolescence is a multi-faceted endeavor requiring change at all levels.

Mitigating e-waste and challenging planned obsolescence requires concerted action from all stakeholders:

- **Consumers** must embrace conscious consumption by prioritizing repair, demanding durable products, and exploring alternatives like Linux to extend the life of their devices.
- **Technology corporations** must move beyond rhetoric and redesign business models for circularity, investing in durability, repairability, and responsible end-of-life management.
- **Governments and policymakers** must enact and enforce robust legislation against planned obsolescence, promote the "right to repair," and invest in e-waste management infrastructure, both domestically and internationally, to prevent the burden from falling disproportionately on developing nations.

The time for incremental change has passed. The approaching obsolescence of Windows 10 serves as a critical wake-up call. This opportunity must be seized to fundamentally rethink the relationship with technology and consumption, transitioning decisively from a linear, wasteful system to a circular, sustainable one. The future of the planet depends on this transformation.

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**Consent to Participate** Not applicable.

**Consent to Publish** Not applicable.

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